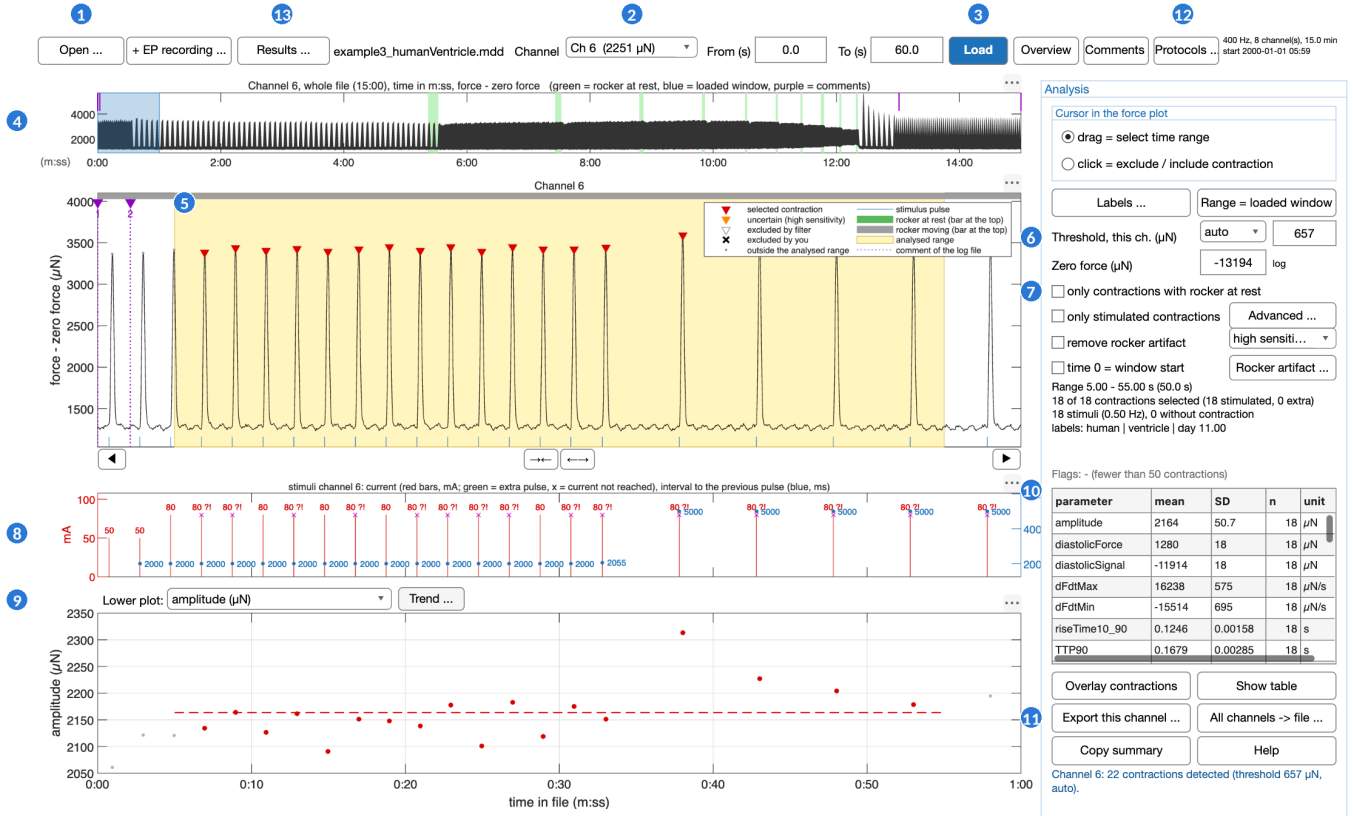
MyoDishAnalysis quick start: the GUI

Version 1.0.0-beta.5 · first analysis, export and the meaning of the results · full help: README, button **Help**, settings guide (docs/MyoDishAnalysis_settings_guide.pdf)

Start. MATLAB (R2019b or newer, no toolboxes) or Python (3.10 or newer); both GUIs look and work the same and give the same results. The log file *<name>_log.log* must be in the folder of the *.mdd* file (sampling rate, calibration, comments).

```
MATLAB:  addpath('/path/to/MyoDishAnalysis');  MyoDishAnalysisGUI      % or MyoDishAnalysisGUI('rec.mdd')
Python:  pip install -e "/path/to/MyoDishAnalysis/py[gui]";  mda-gui   # or mda-gui rec.mdd
```



1	**Open ...** a recording (.mdd): its first 120 s are loaded and analysed. A results file opens an earlier analysis again with its settings and window. The folder is remembered.
2	**Channel**: the list shows the signal range (µN) of each channel: channels with a slice stand out.
3	**From / To** (s) and **Load**: the window to analyse (up to 4 h). Buttons under the force plot (or arrow keys) move and zoom it.
4	**Overview**: the whole recording (green: rocker at rest; purple: comments). Drag in it to load a window; right click: *Select all*. **Comments ...**: numbered marks in the force plot (pointer: text); double-click in the list jumps there.
5	**Force plot**: red triangle: selected, orange: uncertain, grey: excluded by a filter, black ×: by you; blue ticks: stimuli; top bar: rocker (green at rest, grey moving). Drag: analysed range (yellow), shift + drag adds one; right click: zoom to it, all rocker stops side by side. Cursor *click*: exclude / include contractions.
6	**Threshold** per channel: *auto* (default) or a manual value (µN) if noise peaks are counted or small contractions are missed. Zero force: Offset of the log file.
7	**Selection**: only contractions with the rocker at rest; only stimulated contractions; remove the rocker artifact; high sensitivity (counts uncertain contractions) or specificity. **Advanced ...**: all parameters of the method (settings guide); changed ones are counted in red, e.g. *Advanced (3 changed)* (tooltip: which).
8	**Stimuli**: current (mA) and interval (ms) of every pulse; ?! / ×: current not reached.
9	**Lower plot**: one parameter per contraction over time (e.g. run-down of the amplitude). **Trend ...**: whole recording, several channels.
10	**Table**: mean, SD and n of the selected contractions; grey line above: flags of the analysis report for this range. **Show table**: every contraction; **Overlay contractions**: superimposed beats and their mean.
11	**Export this channel ...** (.xlsx or .csv) or **All channels → file** (same range and settings in all channels; with the PDF report if switched on in the menu Save / Export). **Copy summary**: clipboard.
12	**Protocols ...**: protocols found in the log comments (force-frequency, post-rest potentiation, stimulation threshold, refractory period), analysed per frequency, pause, current or S2 interval.
13	**Results ...**: all recordings and channels of a results folder of the watcher with the flags of its report; double-click opens a channel with its settings (red flags: at the flagged window).

Reading the results

What the exported tables contain, how the parameters are defined, and what to check before using them

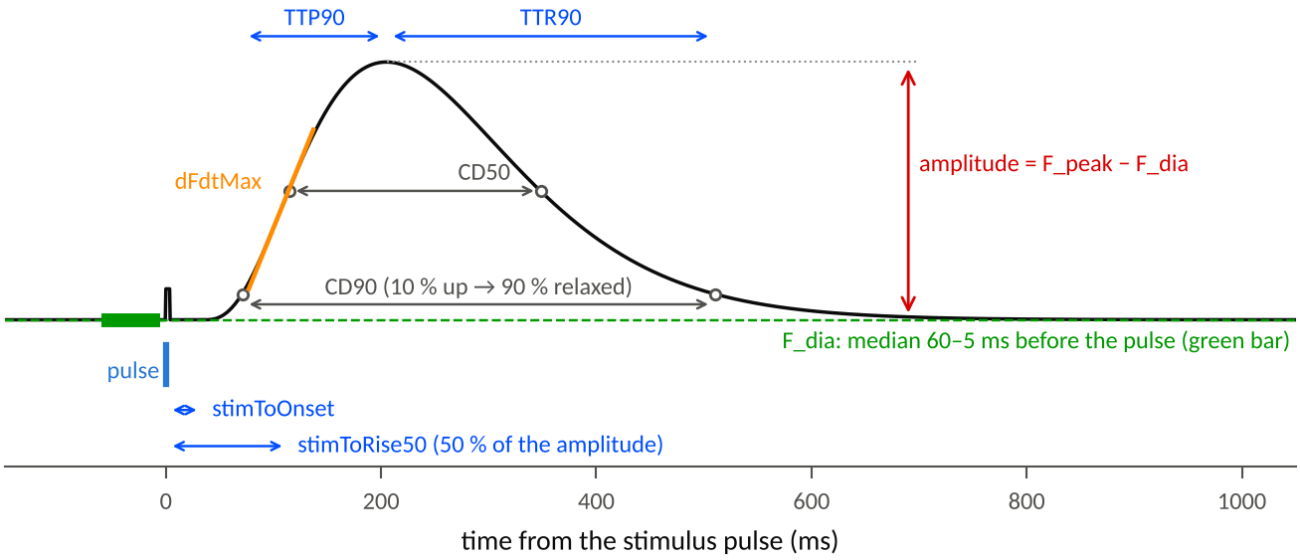


Figure. Parameters of one contraction. Levels in % of the amplitude above F_dia; relaxation levels refer to the minimum after the peak. dFdtMin: fastest relaxation (negative). AUC: area above F_dia from 10 % up to 90 % relaxed. Times in s, forces in µN.

Tables of an export (sheets of the .xlsx file, or <name>_<table>.csv)

table	content
contractions	one row per contraction: time (t_peak, clockTime), beatType, included, uncertain, all parameters, stimulus (t_stim, stimToOnset, stimToRise50, current, interval), rocker state, labels of the channel (species, slice, treatment, …)
summary	per channel and range (or protocol group): counts (nContractions, nStimuli, nMissedBeats, nExtraBeats, nUncertain), capture, and mean, SD, n of every parameter (amplitude_CV)
pulses	every stimulus pulse: outcome (elicited, noResponse, duringContraction), role, contraction
info, parameters	recording, version, date and all settings (an analysis can be repeated or reopened with **Open …**); definitions and units of the parameters
protocolResults	protocols: maximum captured frequency, force-frequency ratios, current thresholds, refractory periods, post-rest potentiation (PRP15 / 30 / 60)

Beat types and counts

term	meaning
stimulated	elicited by a pulse of the channel: the pulse lies up to 0.15 s before the onset (certain contractions: up to 0.5 s, at most half the pacing interval)
extra	no eliciting pulse: spontaneous beat or artifact. **unpaced**: no stimuli in this channel
uncertain	small or not locked to the stimuli; counted only with high sensitivity
included	passes the selection (rocker, stimulated only, not excluded by you): enters the summary
missed beat, capture	pulse without contraction; capture = 1 − missed / stimuli
invertedSignal	force falls after the stimuli (magnet of the MyoDish out of position): rate and capture only, force and time parameters are NaN

Before you use the numbers

- **Look at the markers** in the force plot. Many uncertain contractions together with extra or missed beats (often rocker waves or noise), a long delay (stimToOnset > 150 ms) or an inverted signal: check the threshold and the selection. Uncertain contractions that follow every stimulus are usually real small contractions. The analysis report (PDF) of the command line and the watcher lists these channels.
- **Capture** below 90 %: the slice does not follow the pacing; values of such ranges are not steady-state values.
- **Rocker**: its movement adds a periodic artifact. Compare contractions with the rocker at rest, or remove the artifact (and say so).
- **dF/dt and the times depend on the filter settings** (default: 200 Hz, moving median 50 ms + mean 25 ms). Do not compare values obtained with other settings; the info table records them.
- **diastolicForce** needs a calibrated zero (Offset of the log file); otherwise only changes of diastolicSignal over time are meaningful.
- Use means of ranges with enough contractions (n) and report the version (info table; CITATION.cff).